

CSE101L Quiz 1

Below are the questions for the quiz. Complete each question according to the provided specifications. Ensure that each question's code follows the class and file naming conventions outlined below.

File and Class Naming Conventions

Each question's code should be saved in a file and class as follows:

- **Class Name:** Quiz_{studentID}_QX
- **File Name:** Quiz_{studentID}_QX.java

For example, if your student ID is 12345:

- **Class name for Question 1:** Quiz_12345_Q1
- **File name for Question 1:** Quiz_12345_Q1.java

Question 1: Counting Character Occurrences

Description: The following method, `characterCount`, is provided. Students are required to complete this method so that it counts how many times a specified character appears in the given string. The variable `count` will store the number of occurrences and should be returned at the end.

```
public class Quiz_{studentID}_Q1 {  
  
    public static int characterCount(String text, char character) {  
        int count = 0;  
  
        // Write your code here  
  
        return count;  
    }  
}
```

Parameters:

- `text`: The string in which to search for the character (can be a sentence, word, etc.).
- `character`: The character whose occurrences need to be counted.

Examples:

- If `text = "hello"` and `character = 'l'`, the method should return 2.
- If `text = "apple"` and `character = 'p'`, the method should return 2.
- If `text = "banana"` and `character = 'a'`, the method should return 3.

Details:

- The method is case-sensitive, meaning `'a'` and `'A'` should be treated as distinct characters.
- If the character does not appear in the string, the method should return 0.

Question 2: Expanded Form Representation of a Number

Description: Write the `convertToExpandedForm` method to represent a given number in expanded form, breaking the number into the place value of each digit and displaying them as an addition of terms.

```
public class Quiz_{studentID}_Q2 {  
  
    public static String convertToExpandedForm(int num) {  
        String result = "";  
  
        // Write your code here  
  
        return result;  
    }  
}
```

Examples:

- $12 \rightarrow "10 + 2"$
- $45 \rightarrow "40 + 5"$
- $70304 \rightarrow "70000 + 300 + 4"$

Question 3: Sum of Digit Products

Description: Write a function that takes a number n and returns the sum of the product of the digits of every number from 1 to n . If any number contains a 0, the product of its digits is 0.

```
public class Quiz_{studentID}_Q3 {  
  
    public static int digitProductSum(int n) {  
        int total = 0;  
  
        // Write your code here  
  
        return total;  
    }  
}
```

Examples:

- `digitProductSum(12):`
 - $1 \rightarrow 1$
 - $2 \rightarrow 2$
 - $3 \rightarrow 3$
 - ...
 - $12 \rightarrow 1 * 2 = 2$
 - Sum: $(1 + 2 + 3 + 4 + 5 + 6 + 7 + 8 + 9 + 1 * 0 + 1 * 1 + 1 * 2) = 48$